Advanced SQL
Exercise 2.1
Null functions.

1. Show all employees with their salary. If salary is NULL, display 0.

```
SELECT employee_id,
       name,
   IFNULL(salary, 0) AS salary_with_default
FROM   Employees;
```

Table 1 : Employees

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

2. Show employee names with their department. If department is NULL, show 'Not Assigned'.

```
SELECT employee_id,
       name,
   IFNULL(department, 'Not Assigned') AS department_name
FROM Employees;
```

Table 1: Employees

employee-id	name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

3. Find orders with NULL customer_id using IS NULL().

```
SELECT order_id,
       customer_id
FROM   Orders
WHERE  customer_id IS NULL;
```

Table 2 : orders

order_id	customer_id
103	NULL

4.
```
SELECT order_id,
       customer_id,
   CASE
       WHEN delivery_date IS NULL THEN 'pending'
       ELSE 'Delivered'
   END AS delivery_status
FROM Orders;
```

Table 2 : orders

Order_id	Customer_id	delivery_status
101	201	Delivered
102	202	Pending
103	NULL	Delivered

5. Show all students and their grades Replace NULL with 0.

```
SELECT student_id,
       name,
   IFNULL(grade, 0) AS final_grade
FROM students;
```

Table 3: Students

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

6. Count students who haven't been graded

```
SELECT COUNT(student_id) AS not_graded_count
FROM students
WHERE grade IS NULL;
```

not_graded_count
1

7. Show product name, price, and final price after discount (assume 0 if discount is NULL).

```
SELECT product_id,
       name,
       price - IF NULL(discount, 0) AS final_price
FROM products;
```

product_id	name	final price
501	Keyboard	25
502	Mouse	10
503	Momitor	100

8. Count how many customers have no email

```
SELECT COUNT (customer_id) AS missing_email_count
FROM   customers
WHERE  email IS NULL;
```

missing_email_count
2

9. Show all customers with email. If NULL, display 'No email'.

```
SELECT customer_id,
       name,
    IFNULL(email,'No email') AS email_display
FROM customers;
```

Customer_id	Name	email_display
1	Linda	No email
2	Joseph	joseph@mail.com
3	Nid	No email

10. Show payment details with method replaced by 'unknown' if NULL.

```
SELECT payment_id,
    IFNULL(method, 'unknown') AS method_display,
       status
FROM payments;
```

payment_id	method_display	Status
301	Credit	NULL
302	Paypal	Success
303	Unknown	Failed

11. Show items and their quantity (0 if NULL)

```
SELECT item_id,
       item_name,
       IFNULL(quantity,0) AS quantity_checked
FROM inventory;
```

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

12. Show employee ID and the first available value among bonus or commission

```
SELECT emp_id,
       COALESCE(bonus, commission) AS first_available_reward
FROM employees_extra;
```

emp_id	first_available_reward
1	300
2	100
3	NULL

13. Count classes that dont have a room assigned.

```
SELECT COUNT(class_id) AS no_room_count
FROM   classes
WHERE  room IS NULL;
```

no_room_count
2

14. Show attendance records with status. Replace NULL with 'Not Marked'.

```
SELECT student_id,
       date,
       IFNULL(status, 'Not Marked') AS attendance_status
FROM attendance;
```

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

15. Show account ID, account type (or unknown), and balance (or 0)

```
SELECT account_id,
       IFNULL(account_type, 'unknown') AS type_display,
       IFNULL(balance, 0) AS balance_checked
FROM   Bank_Accounts;
```

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	unknown	2000

16. Show all projects with a start date. If start_date is NULL, display 'TBD'

```
SELECT project_id,
       title,
       IFNULL(start_date, 'TBD') AS start_display
FROM   Projects;
```

project_id	title	start_display
1	Website revamp	2025-01-01
2	Mobile App	TBD
3	Data Migration	TBD

17. Display reviews showing comment (or 'No comment') and rating (or 0)

```
SELECT review_id,
       product_id,
       IFNULL(comment, 'No comment') AS comment_display,
       IFNULL(rating, 0) AS rating_display
FROM   Reviews;
```

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

18. Show the supplier contact number - Use COALESCE(phone, alt_phone, 'No contact')

```
SELECT supplier_id,
       name,
       COALESCE(phone, alt_phone, 'NO CONTACT') AS contact_number
FROM   Suppliers;
```

supplier_id	name	contact_number
1	Global Goods	123456789
2	Best supplies	987654321
3	ValueSource	No contact

19. Show all users and their preferences- Replace all NULLS with defaults :

- Theme - 'Light'
- Language- 'English'
- Timezone - 'UTC'

```
SELECT user_id,
        IFNULL(theme, 'Light') AS theme_set,
        IFNULL(language,'English') A language_set,
        IFNULL(timezone, 'UTC') AS timezone_set
FROM    User_settings;
```

User_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

20. Show maintenance log with:

- issue - default to 'unknown issue'
- technician - default to 'Not Assigned'

```
SELECT record_id,
        machine_id,
        IFNULL(issue, 'unknown issue') AS issue log,
        IFNULL(technician, 'Not Assigned') AS technician_name
FROM    Maintenance;
```

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	uknown issue	Not Assigned
3	M103	Jammed	Alex